

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. Examination (CL)****Apr-May 2018****CL 208 Advanced Surveying****Date: 02/05/2018, Wednesday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q - 1 (a) Do as Directed.****[04]**

- 1) What are the sources of error?
- 2) What is subtense bar?
- 3) The stadia markings are made on
(a) Eyepiece (b) Diaphragm (c) Object Glass (d) none of the above
- 4) An Anallactic lens is provided to make the additive constant equal to
(a) 100 (b) 0 (c) 90

Q - 1 (b) An Angle was observed by the method of repetition by three different observers and the value were as below. **[03]**

Observer No.	Angle A	(weights)
1	$54^{\circ} 30' 20''$	2
2	$54^{\circ} 29' 40''$	2
3	$54^{\circ} 30' 10''$	2

Find most probable value of angle.

Q - 2 (a) Write Any Two.**[14]**

- 1) Describe with example all Laws of Weight.
- 2) Define following terms with examples.

1) Systematic Errors

2) Accidental Errors

3) Observed value

- 4) True value 5) Most Probable Value 6) True error 7) Residual Error.

3) A tacheometer is set up at an intermediate point on a traverse course PQ and the following observations were made on a vertically held staff.

Staff station	Staff Intercept (m)	Vertical Angle	Axial hair Reading (m)
P	2.250 m	$9^{\circ} 30'$	2.105
Q	2.055 m	$6^{\circ} 00'$	1.875

The instrument is fitted with an anallactic lens. The multiplying constant is 100. Compute the length PQ and The RL. Of Q if The R.L. of P is 350.50m.

4) Explain the procedure to determine the tacheometric constants by

- Fixed hair Method
- Movable hair Method

Q - 3 (a) Write Any Two.

[14]

- Derive the distance equation for an anallactic lens. What are its advantages?
- Determine the elevation of top of hilltop using following observation. The height of target was 5 m.

Instrument stations	Staff Reading on B.M	Angle of Elevation	R.L of B.M (m)
P	2.550 m	$18^{\circ} 6'$	345.580
Q	1.670 m	$28^{\circ} 42'$	345.580

Stations P and Q and the hilltop are in the same vertical plane. Find the elevation of the hilltop if the distance between P and Q was 100 m.

- Derive the equation for Base of the object inaccessible and Instrument axes at different levels.
- Enlist the characteristics of tacheometre. Write advantages of tacheometric surveying.

SECTION - II

Q - 4 (a) Briefly describe the equipment's required for aerial photogrammetry. **[07]**

Q - 5 (a) Write any Two. **[14]**

1) Differentiate between:

a. Terrestrial - Aerial photogrammetric

b. Vertical - Oblique photograph

2) Define the following with neat sketch. (Any Two)

Altitude, Declination, Azimuth, The Zenith and Nadir, The prime vertical

3) What do you understand by sidereal day, apparent solar day and mean solar day?

4) Write short note on Any one.

a) Parallax b) procedure for aerial survey c) stereoscope

Q - 6 (a) Write any Two. **[14]**

1) List the objects of geodetic triangulation. Classify triangulation systems with details.

2) Differentiate Plane surveying and geodetic surveying.

3) What do you understand by remote sensing? Briefly explain remote sensing process.

4) Write short note on GPS.
